Southeast Conservation Adaptation Strategy

REVIEW DRAFT: Excerpts from Recent Trends in Southeastern Ecosystems (2024)

Measuring progress toward the Southeast Conservation Adaptation Strategy (SECAS) goal

September 19, 2024

Contents

Executive summary

Through SECAS, diverse partners are working together to design and achieve a connected network of lands and waters that supports thriving fish and wildlife populations and improved quality of life for people across the Southeastern United States and the Caribbean. The long-term goal for SECAS is a 10% or greater improvement in the health, function, and connectivity of Southeastern ecosystems by 2060. To stay on track for achieving that goal, a 1% improvement will be needed every 4 years.

This report is the fifth annual assessment of progress toward the SECAS goal using information from existing monitoring programs. It uses the best available data since SECAS was established in 2011. The report is intended to facilitate discussion around conservation actions needed to meet the goal.

Most indicators improved overall during the period covered in this report. Given the rapid changes happening in the Southeast, this is an encouraging sign for achieving the SECAS goal. Longleaf pine area, prescribed fire in longleaf pine, aquatic connectivity, forested wetland birds, upland hardwood birds, working lands conservation, coastal condition, and marine fisheries indicators improved fast enough to stay on track to meet the SECAS goal. These have all been major areas of shared conservation focus in the Southeast, and those efforts are clearly having a big impact.

Only 9 of the 20 indicators had declining trends. Of these, grassland and savanna birds continue to be the most off track for meeting the SECAS goal. Declines in habitat quantity and quality are likely driving this pattern. There is still hope that focused conservation can have an impact, as targeted improvements in habitat quality in the longleaf pine range resulted in increases in grassland and savanna species like Bachman's sparrow and red-cockaded woodpecker. This further reinforces the importance of accelerating open pine, pine/oak savanna, and other grassland restoration throughout the Southeast for grassland birds, pollinators, and other key species.

To learn more about the role of SECAS in meeting the goal, see the [SECAS Statement of Purpose](#).

Overview of recent trends in ecosystem indicators

Table 1. Overview of recent trends in ecosystem indicators. Indicators shown in green are on track to meet the goal (≥1% increase every 4 years); indicators shown in yellow (<1% increase) and red (declines) are not. Bold indicators were updated in the 2024 report; all others remain unchanged from the previous version.

Ecosystem	Type	Indicator	Yearly % change
Terrestrial			
	Health	Areas without invasive plants	0.33% decline
		Beach birds	**1.42% decline**
		Caribbean undeveloped land	0.39% decline
		Forested wetland area	0.08% increase
		Forested wetland birds	**2.85% increase**
		Gopher tortoise (Eastern population)	Increasing but % change unknown
		Grassland & savanna area	**0.31% decline**
		Grassland & savanna birds	**2.2% decline**
		Longleaf pine area	4.5% increase
		Prescribed fire in longleaf pine	**4.02% increase**
		Salt marsh area	0.03% decline
		Upland forest birds	**0.98% increase**
	Function	Working lands conservation	11% increase
	Connectivity	Landscape condition	0.02% decline
		Undeveloped land in corridors	**0.019% decline**
Freshwater			
	Health	Natural landcover in floodplains	0.008% decline
	Function	Water quality	0.003% increase
	Connectivity	Aquatic connectivity	16% increase
Marine & Estuarine			
	Health	Coastal condition	0.56% increase
	Function	Fisheries	1.1% increase

Introduction

Background

Through SECAS, diverse partners are working together to design and achieve a connected network of lands and waters that supports thriving fish and wildlife populations and improved quality of life for people across the Southeastern United States and the Caribbean. SECAS was started in 2011 by the states of the Southeastern Association of Fish and Wildlife Agencies (SEAFWA) and the federal Southeast Natural Resource Leaders Group. In the fall of 2018, SECAS leadership approved a long-term goal and supporting short-term metrics to evaluate progress toward that connected network.

The long-term goal is a 10% or greater improvement in the health, function, and connectivity of Southeastern ecosystems by 2060. One of the short-term metrics, selected to stay on track to meet the long-term goal, is a 1% improvement in the health, function, and connectivity of Southeastern ecosystems every 4 years. This report on recent trends seeks to measure progress toward that metric.

Purpose of this report

This report assesses progress toward the SECAS goal using information from existing monitoring programs. It is intended to facilitate discussion around conservation actions needed to meet the goal.

Methods

Changes since the last report

For 2024, we made two major improvements: 1) A new indicator for grassland and savanna area, and 2) Updated data and methods for all bird indicators (beach birds, forested wetland birds, grassland and savanna birds, and upland hardwood birds).

Selecting indicators

We selected indicators that are monitored by consistent multi-state efforts and are already used by other organizations to evaluate ecosystem conditions.

Defining health, function, and connectivity

For the purposes of this report, we use these definitions for health, function and connectivity:

- **Health:** The condition of species and the ecosystems they depend on
- **Function:** The benefits provided to people by species and ecosystems
- **Connectivity**: The ability of species and ecosystems to move over time

Defining “recent” trends

SECAS began in 2011, so we focused on trends between 2011-2022. When data were not available for that entire period, we used as much data from that period as possible.

Estimating trends

For indicators where charts only show two points in time (e.g., longleaf pine area), we simply calculated the change between those points. For indicators showing data from more than two years (e.g., prescribed fire in longleaf pine), we estimated the trend based on the slope of a linear regression through all points. For coastal condition, where trends were only available for discrete subregions or states, we averaged trends equally instead of weighting by area.

Evaluating confidence in trend

We used a combination of quantitative methods and qualitative judgement to estimate confidence in the trend. For indicators where the trend is a regression, we used the p-value to estimate the likelihood that the trend is non-zero. We considered any trend with a p-value > 0.10 as not significant and having low confidence. For significant trends, we used qualitative judgement based on the design of the monitoring, overall sample size, and major sources of variability in the indicator to determine whether the confidence is medium or high. For indicators where the trend is not based on a regression, we only used the qualitative judgement.

Assessments used in the report

We used 13 different assessments to evaluate indicator trends. Assessments ranged from remotely sensed data like the National Land Cover Database to long-term volunteer-driven monitoring programs like eBird. Additional assessments used included America’s Longleaf Range-wide Accomplishment Reports, Forest Inventory and Analysis, Gopher Tortoise Candidate Conservation Agreement reports, USDA Soil and Water Resources Conservation Act Reports, Southeast Conservation Blueprint, EPA 303(d) state reports, Southeast Aquatic Resources Partnership Aquatic Barrier Database, NOAA C-CAP Regional Landcover, National Coastal Condition Assessments, NOAA Reports to Congress on the Status of Fisheries, and Esri global landcover.

Assessments considered but not used in this report

There are many subregional assessments of ecosystem conditions (e.g., Chesapeake Bay, Everglades), but their coverage of only part of the Southeast made them difficult to formally integrate into this particular report. One national assessment, Surfrider Foundation’s State of the Beach, had potential, but was not used because it focused on policies related to beach conditions rather than the actual condition of the beaches.

Ecosystem indicator trends

Terrestrial

All inland and coastal terrestrial ecosystems

Health

The condition of species and the ecosystems they depend on

Beach birds

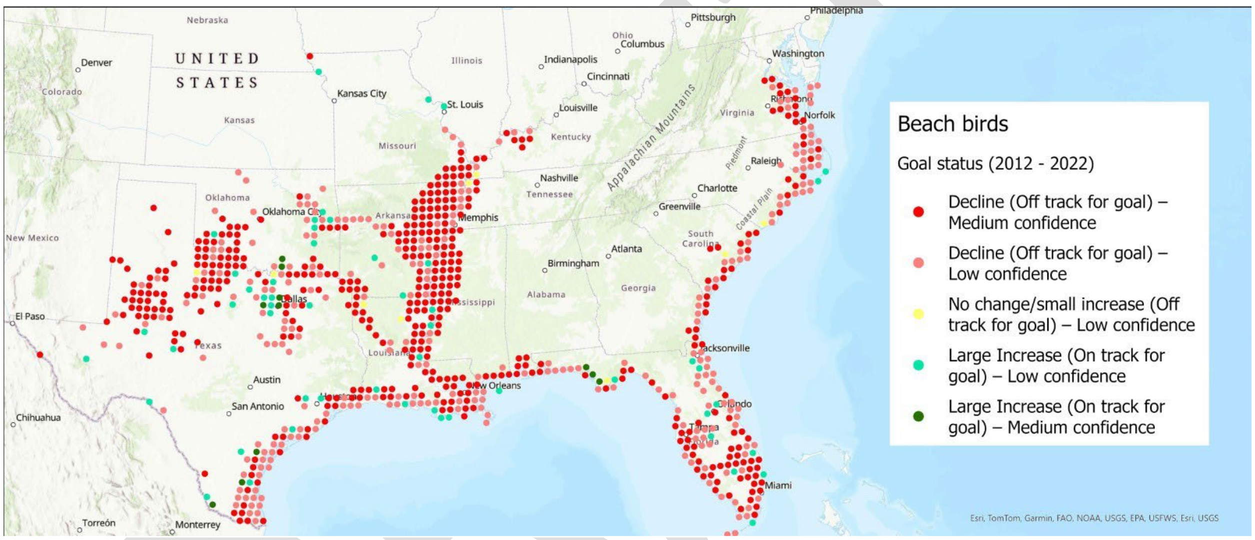


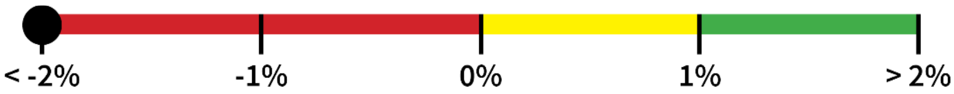
Figure 1. Trends in beach bird abundance from 2012-2022. Note: This indicator also includes interior least tern, which uses beach habitat far from the coast.

Yearly trend

When averaged across all points with trends, beach bird abundance decreased by 1.42% per year from 2012-2022. Species used were American oystercatcher, black skimmer, gull-billed tern, least tern, piping plover, roseate tern, snowy plover, and willet. These species are Regional Species of Greatest Conservation Need for states in the Southeast, primarily occur in this ecosystem, and have sufficient data for trend analysis in eBird Status and Trends. Most points across the SECAS region were declining, with the exception of some areas with significant investment in coastal conservation. Individual species trends also followed this pattern. The only exception was breeding piping plover, where all breeding points in the SECAS region were declining.

On track to meet SECAS goal

No. The decline of about 5.68% every 4 years is not enough to reach the SECAS goal of a 1% increase every 4 years.



Data source

[eBird Status and Trends]

Confidence in trend

Medium. Most points that had declines (52%) were statistically significant.

Interpretation

This is an indicator of beach habitat quality. The mixed but mostly declining trends highlight the challenges and opportunities within this ecosystem. Habitat modification, climate change, and human disturbance continue to pose problems, but conservation action in specific areas seems to be making an impact. As these birds are migratory, conservation actions and threats impacting their populations occur both within the Southeast and in other parts of the species' ranges.

Forested wetland birds

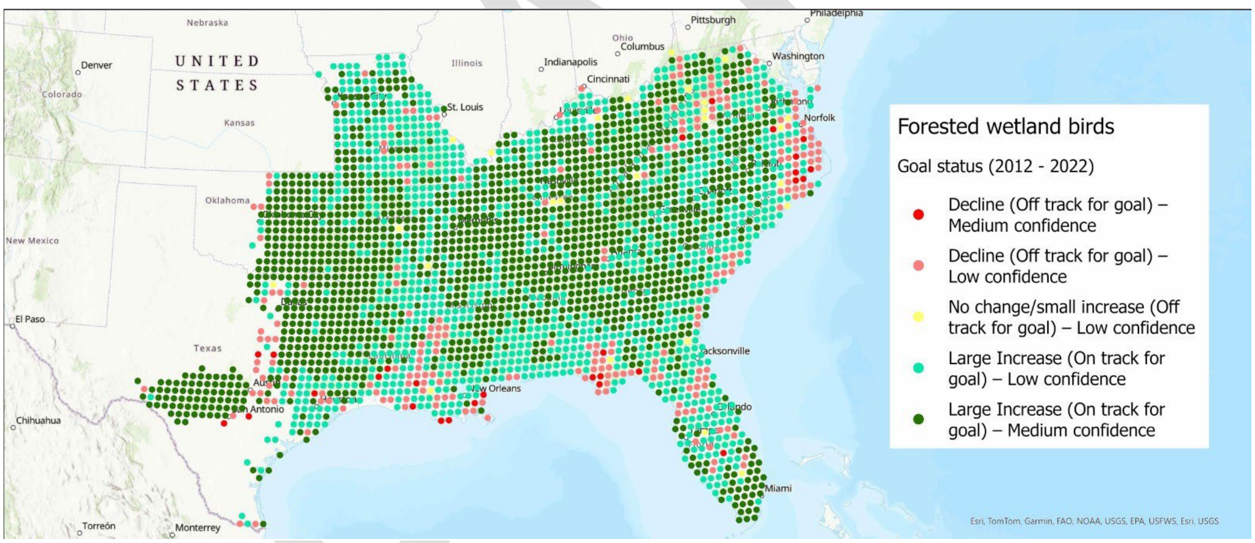


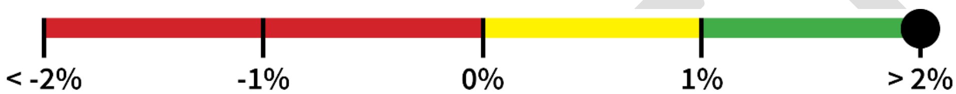
Figure 2. Trends in forested wetland bird abundance from 2012-2022.

Yearly trend

When averaged across all points with trends, forest wetland bird abundance increased by 2.85% per year from 2012-2022. Species used were prothonotary warbler, Swainson's warbler, swallow-tailed kite, and yellow-throated warbler. These species are are Regional Species of Greatest Conservation Need for states in the Southeast, primarily occur in this ecosystem, and have sufficient data for trend analysis in eBird Status and Trends. Most points across the SECAS region were increasing. Declines were mostly in areas experiencing major impacts from sea-level rise. Individual species trends also followed this pattern.

On track to meet SECAS goal

Yes. The increase of about 11.4% every 4 years is greater than the SECAS goal of a 1% increase every 4 years.



Data source

[eBird Status and Trends]

Confidence in trend

Medium. Most of the points (57%) that were on track for the goal were statistically significant.

Interpretation

This is an indicator of both local and landscape conditions across the forested wetland ecosystem. While there are some declines, especially in areas impacted by sea-level rise, overall, forested wetland birds appear to be on track to meet the SECAS goal. This may be due to the extensive conservation investments in forested wetlands, policies restricting wetland development, and growing interest from urban communities in protecting water supply and reducing flood risks.

Grassland & savanna area

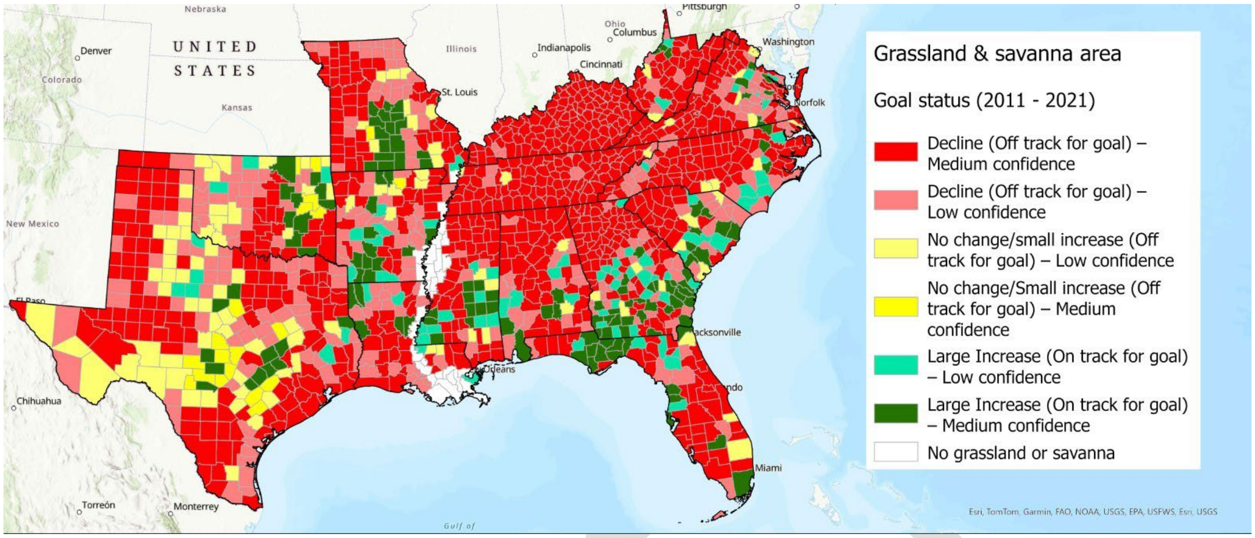


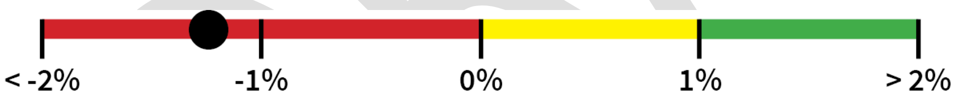
Figure 3. Grassland and savanna area trends from 2011 – 2021.

Yearly trend

Grassland and savanna area declined by 0.31% per year from 2011-2021. Area declined in most counties (76%). Increases typically occurred in counties with major ecosystem-based restoration efforts (e.g., range-wide longleaf pine, woodlands in Missouri and Arkansas.)

On track to meet SECAS goal

No. The decline of about 1.24% every 4 years is not enough to reach the SECAS goal of a 1% increase every 4 years.



Data source

Modeled indicator based on [National Land Cover Database (NLCD)]() land cover and canopy cover, [LANDFIRE Biophysical Settings,]() [Texas Ecological Mapping Systems,]() [Oklahoma Ecological Systems Map.]() The model starts with areas that were historically grassland or savanna based on LANDFIRE. It then uses landcover and canopy cover to identify current area for each year. For areas of historic grassland, it uses a canopy cover threshold of 0%. For areas of historic savanna, it uses a canopy cover threshold of 60%. In Texas and Oklahoma, we further reduced the estimate by removing mesquite-invaded areas that weren't currently functioning as grassland or savanna. This approach is similar to parts of the grasslands and savannas Blueprint indicator, but includes a number of changes more focused on trend estimation.

Confidence in trend

High. Trend is statistically significant and shows consistent declines across all five years with data (2011, 2013, 2016, 2019, 2021).

Interpretation

This is a coarse indicator of the overall extent of grassland and savanna. It includes a wide range of quality, from restored areas and remnants, to temporary grasslands created by forestry, to highly altered areas of pasture. The steep declines mirror large declines seen in species that depend on grasslands and savannas, like pollinators and grassland birds. Grassland declines throughout the SECAS region are occurring on both public and private lands.

For most SECAS states, the major source of grassland and savanna loss was excess tree growth. The exceptions to this were: 1) Missouri, where the biggest loss was to row crop, and 2) Texas and Florida, where the biggest loss was to urban growth. Despite steep declines, improvements in places like the longleaf pine range, historic woodlands in Missouri and Arkansas, and tallgrass prairie in Northeast Oklahoma show that focused conservation attention can reverse declines in specific places.

Grassland & savanna birds

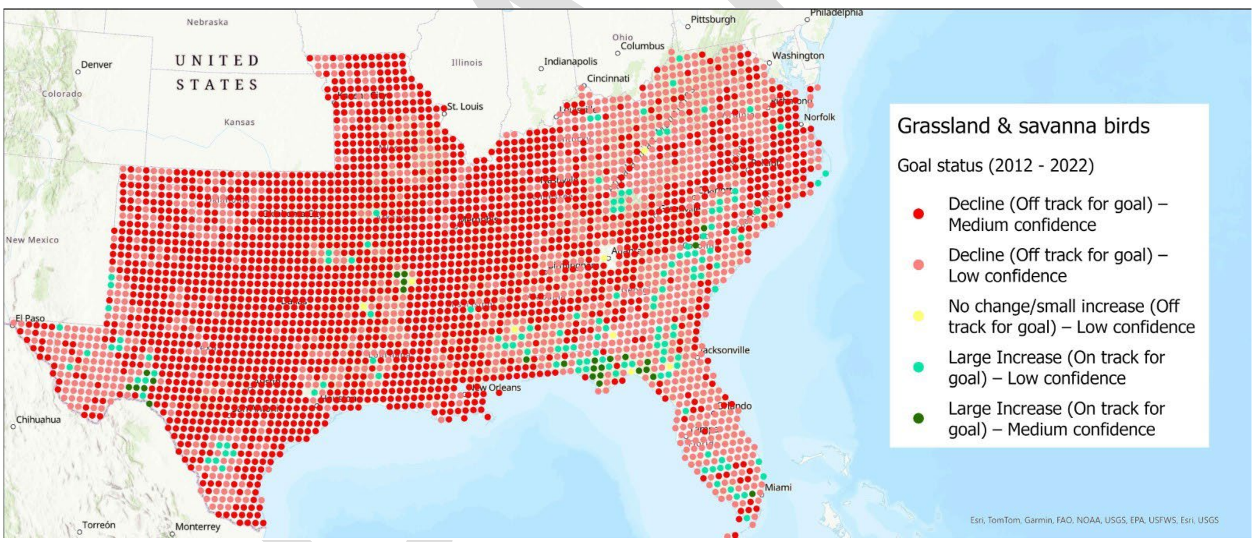


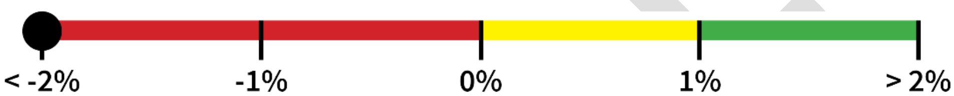
Figure 4. Trends in grassland and savanna bird abundance from 2012-2022.

Yearly trend

When averaged across all points with trends, grassland and savanna bird abundance declined by 2.2% per year from 2012-2022. Species used were American kestrel, Bachman's sparrow, Eastern meadowlark, grasshopper sparrow, Henslow's sparrow, LeConte's sparrow, loggerhead shrike, Northern Bobwhite, prairie warbler, red-cockaded woodpecker, and scissor-tailed flycatcher. These species are Regional Species of Greatest Conservation Need for states in the Southeast, primarily occur in this ecosystem, and have sufficient data for trend analysis in eBird Status and Trends. Most points across the SECAS region were declining. For most individual species, a majority of points were declining, but there were also a number of points with increases. Two species with most of their range in the longleaf pine ecosystem had a larger number of increasing points than other species: red-cockaded woodpecker (58% increasing) and Bachman’s sparrow (43% increasing).

On track to meet SECAS goal

No. The decline of about 8.8% every 4 years is not enough to meet the SECAS goal of a 1% increase every 4 years.



Data source

[eBird Status and Trends]

Confidence in trend

Medium. Most of the points (65%) that were declining and off track for the goal were statistically significant.

Interpretation

This is an indicator of both local and landscape conditions across the grassland and savanna ecosystem. Large declines across most of the region highlight the major problems for this ecosystem and the species that depend on it. Signs of improvement in the longleaf range, South Florida, the Chihuahuan Desert, the West Gulf Coastal Plain, and the Appalachians show that targeted conservation attention can still have an impact. Improvements in specific species like red-cockaded woodpecker and Bachman’s sparrow also show that targeted improvements in habitat quality can make a major difference.

Prescribed fire in longleaf pine

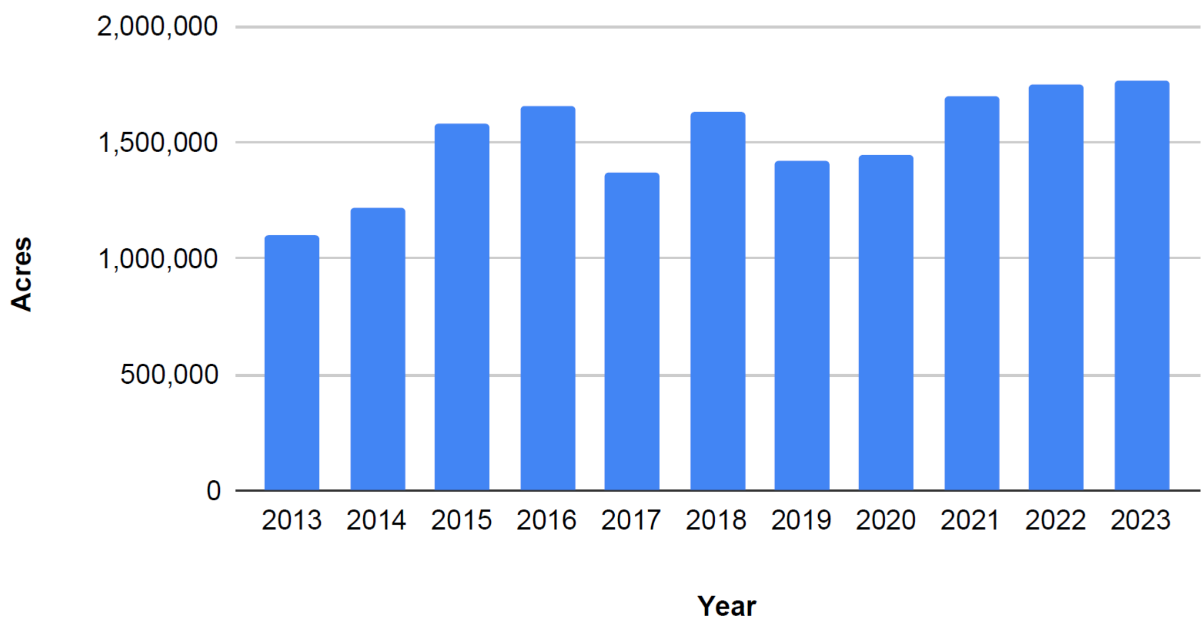


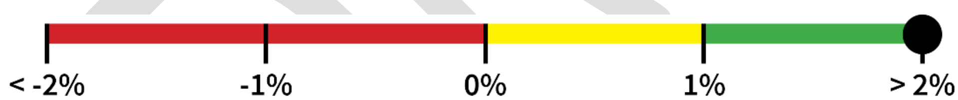
Figure 5. Acres of prescribed fire in longleaf pine from 2013-2023.

Yearly trend

Prescribed fire in longleaf pine increased by about 4.02% per year from 2013-2023.

On track to meet SECAS goal

Yes. The increase of about 16% every 4 years is greater than the SECAS goal of a 1% increase every 4 years.



Data source

[America's Longleaf Range-wide Accomplishment Reports]

Confidence in trend

Medium. The trend is statistically significant. While the range-wide tracking system for prescribed fire in longleaf is not perfect, its strong coverage of significant geographic areas means it likely documents a large percentage of prescribed fire in longleaf over this period.

Interpretation

This is an indicator of habitat management in one part of the grassland and savanna ecosystem. Prescribed fire is also important outside of the longleaf range, but sufficient trend data wasn't yet available in those areas. After a dip in prescribed fire in 2019 and 2020, numbers significantly increased in 2021,2022, and 2023—setting new records in all three years. This suggests that collaborative longleaf restoration efforts are continuing to improve the condition of this important ecosystem.

Upland forest birds

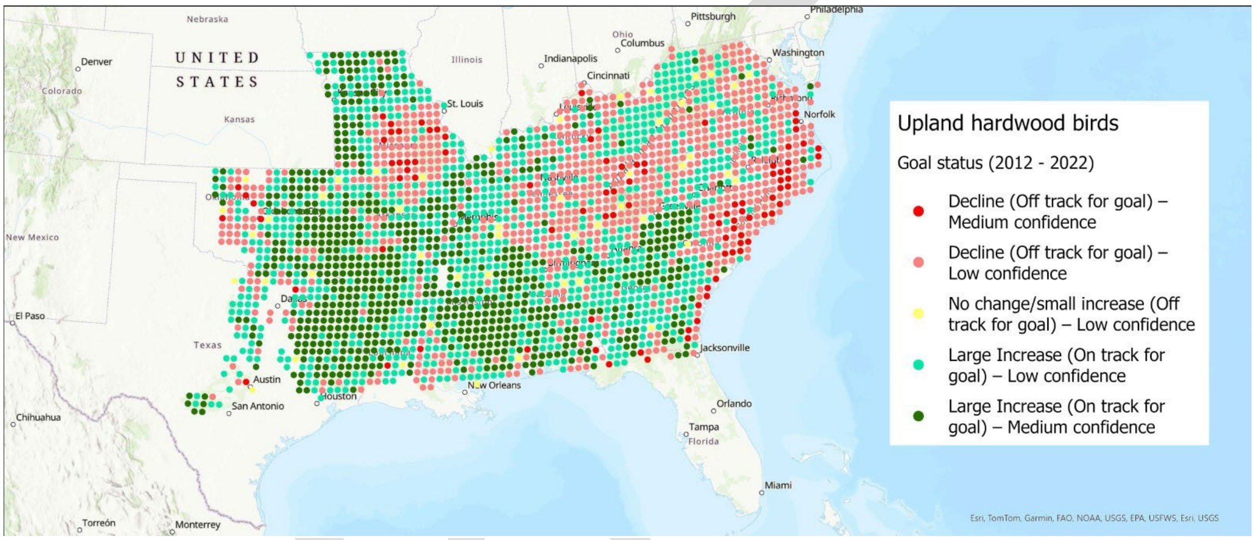


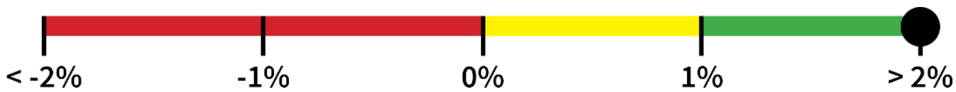
Figure 6. Trends in upland hardwood birds from 2012 - 2022.

Yearly trend

When averaged across all points with trends, upland forest bird abundance increased by 0.98% per year from 2012-2022. Species used were cerulean warbler, Louisiana waterthrush wood thrush, worm-eating warbler. These species are Regional Species of Greatest Conservation Need for states in the Southeast, primarily occur in this ecosystem, and have sufficient data for trend analysis in eBird Status and Trends. Trends varied across the Southeast, with the biggest declines occurring in the Central Hardwoods, Appalachians, and northeast part of the Southeast Coastal Plain Bird Conservation Regions. For two widespread species, points were mostly increasing: wood thrush (91% increasing) and Louisiana waterthrush (64% increasing). For two species with smaller ranges in the Southeast, points were mostly declining: cerulean warbler (78% declining) and worm-eating warbler (78% declining).

On track to meet SECAS goal

Yes. The increase of about 3.92% every 4 years is greater than the SECAS goal of a 1% increase every 4 years.



Data source

[eBird Status and Trends]

Confidence in trend

Low. Less than half of the points that were increasing (38%) were statistically significant.

Interpretation

This is an indicator of both local and landscape conditions across the upland forest ecosystem. Upland hardwood birds benefit from conversion of historic grassland and savanna ecosystems into closed canopy forest. In areas with increases, that may mean increases in closed canopy forest overall are offsetting the negative impacts of land use changes like greater forest fragmentation. Some areas of upland forest bird decline, like Southeast Missouri, could actually be positive signs of conservation overall as these areas are restored to the more open forest types that historically occurred there.

Species-specific trends also highlight how more widespread generalist species (Louisiana waterthrush, wood thrush) seem to be poised to take advantage of changing landscape conditions. More specialist and range-limited species (cerulean warbler, worm-eating warbler) seem to be less able to take advantage of these changes. Based on range-wide trends for these species, it doesn’t appear that climate change is a major driver of trends during this time period. It’s also important to note that all these species are neotropical migrants. Threats to survival during migration (e.g., communication towers) and on their wintering grounds (e.g., habitat loss) are likely also impacting population trends.

Connectivity

The ability of species and ecosystems to move over time

Undeveloped land in corridors

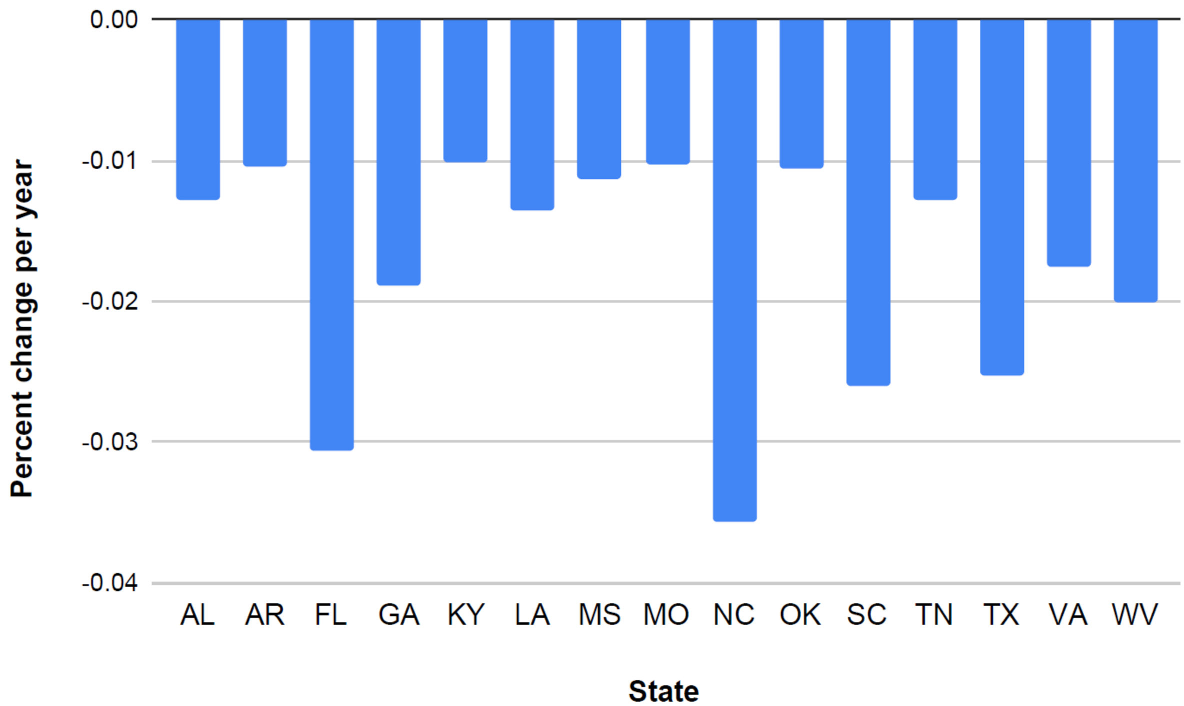


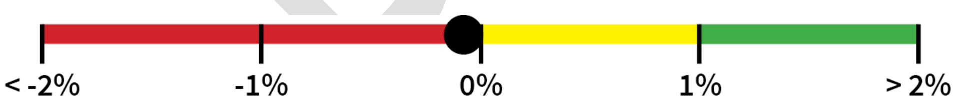
Figure 7. Percent change per year in undeveloped land within Southeast Conservation Blueprint 2024 continental corridors from 2011-2021.Suffient landcover data were not available to include Puerto Rico and the U.S. Virgin Islands in this analysis.

Yearly trend

All states showed statistically significant declining trends for undeveloped land in corridors. Undeveloped area within corridors across all SECAS states combined declined by 0.019% per year.

On track to meet SECAS goal

No. The decline of about 0.076% every 4 years is not enough to reach the SECAS goal of a 1% increase every 4 years.



Data source

[National Land Cover Database (NLCD)] and [Southeast Conservation Blueprint 2024 continental corridors]

Confidence in trend

High. The National Land Cover Database (NLCD) is particularly good at separating developed and not developed areas. There were also enough years available to get statistically significant trends for each state and all states combined.

Interpretation

This is an indicator of terrestrial connectivity that looks at the landcover change within the Southeast Blueprint 2024 continental corridors. It does not account for changes in the Blueprint corridor layer across different versions of the Blueprint. Development pressure varies across the states of the Southeast, so having undeveloped areas in corridors is easier for some states than others. When looking at population growth rates during the general period this indicator covers, some state performed as expected based on their population growth while others didn't. States with the largest indicator declines like North Carolina, Florida, South Carolina, and Texas also had relatively high population growth rates. One interesting exception to this was West Virginia, which had the fifth highest decline in undeveloped land in corridors despite being only one of two SECAS states with a negative population growth rate.